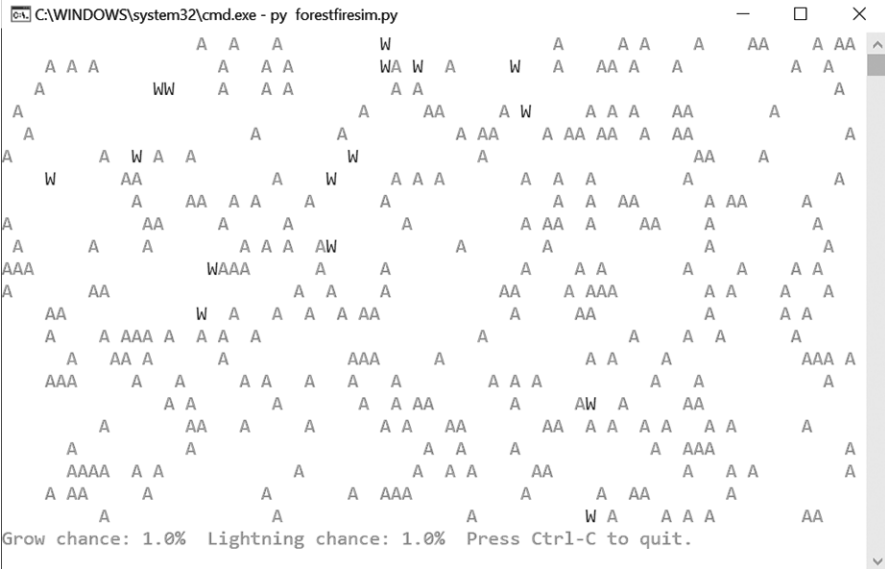


The Program in Action

When you run *forestfiresim.py*, the output will look like this:



```
C:\WINDOWS\system32\cmd.exe - py forestfiresim.py
      A A A      A A A      W      A      A A A      AA      A AA
    A A      WW      A A A      A A      A      AA A      A A
  A      A      A      A      A      A W      A A A      AA      A
A      A      A      A      A      A AA      A AA AA      A AA      A
W      A W A      A      W      A A A      A      A      A      A
  A      A      AA      A A      A      A      A      A      A AA      A
A      A      AA      A      A      A      A AA      A      AA      A
A      A      A      A A A      A      A      A AA      A      A
AAA      A      WAAA      A      A      A      A      A      A A      A
A      AA      W      A      A A      A      AA      A      A      A
  A      A AAA A      A A      A      A      A      A      A      A
    A      AA A      A      AAA      A      A      A      A      A
AAA      A      A A      A      A      A      A      A      A      A
      A      AA      A      A      A      A      A      A      A
    AAAA      A      A      A      A      A      A      A      A
  A AA      A      A      A      A      A      A      A      A
      A      A      A      A      A      A      A      A      A
      A      A      A      A      A      A      A      A      A
      W A      A A A      AA
Grow chance: 1.0% Lightning chance: 1.0% Press Ctrl-C to quit.
```

Figure 29-1: The forest fire simulation, with green As for trees and red Ws for flames

How It Works

This simulation is an example of *emergent behavior*—the interaction between simple parts in a system creating complicated patterns. Empty spaces grow into trees, lightning turns trees into fire, and fire turns trees back into empty spaces while spreading to neighboring trees. By adjusting the tree growth and lightning strike rate, you can cause the forest to display different phenomena. For example, a low lightning chance but high growth rate causes large, constant forest fires, since the trees tend to be near each other and quickly replenish. A low growth rate but high lightning strike chance creates several small fires that quickly extinguish due to a lack of nearby trees. We don't explicitly program any of this behavior; rather, it naturally emerges from the system that we created.

1. `"""Forest Fire Sim, by Al Sweigart al@inventwithpython.com`
2. `A simulation of wildfires spreading in a forest. Press Ctrl-C to stop.`
3. `Inspired by Nicky Case's Emoji Sim http://ncase.me/simulating/model/`
4. `View this code at https://nostarch.com/big-book-small-python-projects`
5. `Tags: short, bext, simulation"""`
6.
7. `import random, sys, time`
8.
9. `try:`
10. `import bext`